

Project Initialization and Planning Phase Project Proposal (Proposed Solution)

Date	7 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Report

Smart Lender transforms loan approval decisions using machine learning classification models, boosting efficiency and accuracy in creditworthiness assessment. It tackles the inefficiencies of manual, rule-based screening, promising faster operations, reduced credit risk, and better outcomes for both institutions and applicants. Key features include a machine learning-based credit assessment model (XGBoost), real-time decision-making, and batch evaluation for high-volume processing.

Project Overview	
Objective	Predict the creditworthiness of loan applicants using machine learning classification models (Decision Tree, Random Forest, K-Nearest Neighbors, XGBoost), enabling financial institutions to make faster, more consistent, and data-driven loan approval decisions.
Scope	Covers data preprocessing of structured applicant attributes (gender, marital status, education, employment status, income, loan amount, loan term, credit history, property area); training and comparing four classification algorithms; selecting the best-performing model; and deploying it in a Flask web application supporting both real-time single-applicant prediction and batch evaluation, with a deployment plan on IBM Cloud.
Problem Statement	
Description	Traditional loan approval processes are slow, manual, and inconsistent, especially at high application volumes. Human assessment can overlook patterns hidden in historical data, and rule-based screening fails to adapt to complex applicant profiles.
Impact	Slow, inconsistent screening increases operational workload, delays decisions for applicants, and raises the risk of approving high-risk applicants or rejecting creditworthy ones — ultimately affecting non-performing assets, customer satisfaction, and institutional efficiency.
Proposed Solution	
Approach	Preprocess structured applicant data (handle missing values, encode categorical variables, scale numerical features), then train and evaluate Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost classifiers using accuracy, precision, recall, F1-score, and confusion matrix. XGBoost is selected as the best-performing model (94.7% training accuracy, 81.1% testing accuracy) and integrated into a Flask web application for real-time and batch predictions, deployable on IBM Cloud Code Engine.
Key Features	- Machine learning-based credit assessment model (XGBoost) benchmarked against Decision Tree, Random Forest, and KNN. - Real-time single-applicant prediction with approval decision, likelihood score, and risk label (Low/Medium/High). - Fast-Track Approval scenario for low-risk applicants with stable income and positive credit history. - High-Risk Detection scenario flagging irregular income, no credit history, or high loan burden for manual review. - Batch evaluation via CSV/Excel upload for loan officers processing multiple applicants at once. - Cloud-ready deployment on IBM Cloud Code Engine using a containerized Flask backend.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	T4 GPU
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, xgboost
Development Environment	IDE	Jupyter Notebook, PyCharm
Data		
Data	Source, size, format	Kaggle dataset, 614 records, CSV UCI dataset, 690 records, CSV